

Bundle Structures in Topology

Labix

March 15, 2026

Abstract

- Notes on Algebraic Topology by Oscar Randal-Williams

Contents

1	Vector Bundles	3
1.1	Basic Definitions	3
1.2	The Cocycle Conditions	5
1.3	Operations on Vector Bundles	6
1.4	Homotopy Theory of Vector Bundles	7
1.5	Almost Complex Structures	8
2	Sequences of Vector Bundles	10
2.1	Subbundles and Quotient Bundles	10
2.2	The Kernel of a Morphism	10
2.3	Short Exact Sequences of Vector Bundles	10
3	Reduction of Structure Group	12
3.1	Structure Groups	12
3.2	Metrics and Reduction to Orthogonal Groups	12
3.3	Orientability and Reduction to Special Linear Groups	12
3.4	Subbundles and Reduction to Product Orthogonal Subgroups	12
4	The Topology of Fiber Bundles	13
4.1	Fiber Bundles	13
4.2	Sections of a Bundle	15
4.3	Sphere Bundles	15

1 Vector Bundles

1.1 Basic Definitions

Definition 1.1.1 (Vector Bundles)

Let F be a field. Let E, B be spaces with B connected. Let $p : E \rightarrow B$ be a continuous map. We say that p is a vector bundle of dimension $d \in \mathbb{N}$ over F if the following are true.

- $p : E \rightarrow B$ is surjective.
- Local Triviality: There exists a collection of homeomorphisms

$$\{(U_i \subseteq B, \phi_i : p^{-1}(U_i) \rightarrow U_i \times F^k) \mid i \in I\}$$

called local trivialization charts such that $B = \bigcup_{i \in I} U_i$ and the following diagram commutes:

$$\begin{array}{ccc} p^{-1}(U_i) & \xrightarrow{\phi_i} & U_i \times F^k \\ & \searrow p \quad \swarrow \pi & \\ & U_i & \end{array}$$

where π is the projection by forgetting the second variable.

We say that B is the base space, E the total space. It is denoted as (F, E, B) .

The local trivialization means that locally at a neighbourhood, the vector bundle looks the same the open set times F^k . In particular, there is also a notion of trivial bundle which means that the bundle is globally just $B \times \mathbb{R}^r$.

Definition 1.1.2 (The Fiber of a Bundle)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Let $b \in B$. Define the fiber of E at B to be

$$E_b = p^{-1}(b)$$

Example 1.1.3 (The Trivial Bundle)

Let B be a space. Let F be a field. The trivial bundle $p : B \times F^k \rightarrow B$ defined by $p(b, v) = b$ is a vector bundle. The bundle is denoted by e^k .

Example 1.1.4 (The Mobius Strip)

Let $E = \frac{I \times \mathbb{R}}{\sim}$ where $(0, x) \sim (1, -x)$ for $x \in \mathbb{R}$. Then the projection map $E \rightarrow \frac{I}{0 \sim 1} \cong S^1$ is a vector bundle of rank 1.

Example 1.1.5 (The Real Line Bundle over $\mathbb{RP}^n$)

Consider $\mathbb{RP}^n = \frac{\mathbb{R}^{n+1} \setminus \{0\}}{\sim}$ where $x \sim y$ if $y = \lambda x$ for some $\lambda \in \mathbb{R}^\times$. Define the line bundle

$$\gamma_1 : E_{\gamma_1} = \{([x], v) \in \mathbb{RP}^n \times \mathbb{R}^{n+1} \mid v \in [x]\} \rightarrow \mathbb{RP}^n$$

to be the map $([x], v) \mapsto [x]$ where $v \in [x]$ or $v = 0$. The local trivialization charts are given by the open cover $U_0, \dots, U_n$ where

$$U_i = \{[x_0 : \dots : x_n] \mid x_i \neq 0\}$$

and the map $\gamma_1^{-1}(U_i) \rightarrow U_i \times \mathbb{R}$ is given by $([x], v) \mapsto ([x], \lambda)$ where $v = \lambda(x_0, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_n)$ and $[x] = [x_0 : \dots : x_{i-1} : 1 : x_{i+1} : \dots : x_n]$.

Example 1.1.6 (The Complex Line Bundle over $\mathbb{CP}^n$)

Consider $\mathbb{CP}^n = \frac{\mathbb{C}^{n+1} \setminus \{0\}}{\sim}$ where $x \sim y$ if $y = \lambda x$ for some $\lambda \in \mathbb{C}^\times$. Define the line bundle $\gamma_1 : \{([x], v) \in \mathbb{CP}^n \times \mathbb{C}^{n+1} \mid v \in [x]\} \rightarrow \mathbb{CP}^n$ to be the map $([x], v) \mapsto [x]$ where $v \in [x]$ or $v = 0$.

Lemma 1.1.7

Let F be a field. Let $p : E \rightarrow B$ be a vector bundle of dimension $d \in \mathbb{N}$ over F . Then the local trivialization charts induces a homeomorphism

$$E_b \cong F^d$$

for all $b \in B$.

Definition 1.1.8 (Morphism of Vector Bundles)

Let $p_1 : E_1 \rightarrow B_1$ and $p_2 : E_2 \rightarrow B_2$ be vector bundles. A morphism of these vector bundles is given by a pair of continuous maps $f : E_1 \rightarrow E_2$ and $g : B_1 \rightarrow B_2$ such that the following diagram commutes

$$\begin{array}{ccc} E_1 & \xrightarrow{f} & E_2 \\ \downarrow p_1 & & \downarrow p_2 \\ B_1 & \xrightarrow{g} & B_2 \end{array}$$

and $f|_{p_1^{-1}(b)}$ is a linear map for all $b \in B$.

Definition 1.1.9 (Isomorphism of Vector Bundles) A bundle homomorphism from E_1 to E_2 is an isomorphism if there exists an inverse bundle homomorphism from E_2 to E_1 . In this case, we say that E_1 and E_2 are isomorphic.

Lemma 1.1.10

Let B be space. Let $p_1 : E_1 \rightarrow B$ and $p_2 : E_2 \rightarrow B$ be vector bundles. Let $f : E_1 \rightarrow E_2$ be a morphism of vector bundles. Then f is an isomorphism if and only if $f|_{p_1^{-1}(b)}$ is an isomorphism for all $b \in B$.

Proof

If f is an isomorphism, then the inverse of $f|_{p_1^{-1}(b)}$ is clearly given by $f^{-1}|_{p_2^{-1}(b)}$.

Now suppose that $f|_{p_1^{-1}(b)}$ is an isomorphism for all $b \in B$. Then clearly f is bijective. I claim that $p_2 = p_1 \circ f^{-1}$. Let $x \in E_2$. Then there exists $y \in E_1$ such that $f(y) = x$. Then $p_1(f^{-1}(x)) = p_1(y)$. But $p_1 = p_2 \circ f$ by definition of morphisms of vector bundles. Hence $p_1(y) = p_2(f(y)) = p_2(x)$ and we conclude.

It remains to show that f^{-1} is continuous. Let $U \subseteq E_1$ be an open set. Then there exists local trivialization charts (V_i, ϕ_i) of E_1 such that the V_i 's cover $p_1(U)$. Then we have $U = \bigcup_{i \in I} U \cap V_i$ and $f^{-1}(U)$ is open if and only if $f^{-1}(U \cap V_i)$ is open. But since restrictions of local trivialization charts are also local trivialization charts, it suffices to prove that f is continuous on local trivialization charts.

So let (U, ϕ) be a local trivialization chart of E_1 . Since E_2 is also covered by local trivialization charts, WLOG assume that (U, ψ) is a local trivialization chart of E_2 . Now the map $\psi \circ f \circ \phi^{-1} : U \times F^k \rightarrow U \times F^n$ sends (x, v) to $(x, g(x)(v))$ for some function $g(x) : F^k \rightarrow F^n$. Since the map is a homeomorphism, the map $v \mapsto g(x)(v)$ must be a homeomorphism, and hence $k = n$. Since restriction to each fiber is a bijection, $g(x)$ must be an invertible linear map and hence continuous. Hence $\phi \circ f^{-1} \circ \psi^{-1}$ is continuous. Thus f is continuous. ■

Definition 1.1.11 (Sections)

Let k be a field. Let $p : E \rightarrow B$ be a vector bundle over k . A section of p is a map $s : B \rightarrow E$ such that $p \circ s = \text{id}_B$.

Lemma 1.1.12

Let k be a field. Let $p : E \rightarrow B$ be a vector bundle over k . Then the following are true.

- Let $s_1, s_2 : B \rightarrow E$ be sections of p . Then $s_1 + s_2$ is a section of p .
- Let $s : B \rightarrow E$ be a section of p . Let $\lambda \in k$. Then λs is a section of p .

Definition 1.1.13 (The Space of Sections)

Let F be a field. Let $p : E \rightarrow B$ be a vector bundle over F . Define the F -vector space of sections to be the set

$$s(E) = \{s : B \rightarrow E \mid s \text{ is a section of } p\}$$

together with addition and scalar multiplication defined pointwise.

Lemma 1.1.14

Let F be a field. Let $p : E \rightarrow B$ be a vector bundle over F . Then E is isomorphic to the trivial bundle if and only if there exists global sections $s_1, \dots, s_n \in s(E)$ such that $s_1(b), \dots, s_n(b)$ is a basis for $p^{-1}(b)$ for all $b \in B$.

Example 1.1.15

The Mobius vector bundle $M \rightarrow S^1$ is non-trivial.

Proof

Suppose that it is trivial. By the above lemma, there exists a global section $s : S^1 \rightarrow M$ such that $s(e^{2\pi it}) \neq [(t, 0)]$ for all $t \in [0, 1]$. Writing $M = \frac{I \times \mathbb{R}}{\sim}$ where $(0, x) \sim (1, -x)$, we see that the map s is of the form $s(e^{2\pi it}) = [(t, f(t))]$ for some continuous map $f : I \rightarrow \mathbb{R}$ such that $f(0) = -f(1)$ by definition of the equivalence relation. But by the intermediate value theorem, there exists $c \in I$ such that $f(c) = 0$, contradicting the fact that s is nowhere vanishing. ■

Definition 1.1.16 (Set of Isomorphism Classes of Vector Bundles)

Let X be a space. Let F be a field. Denote the set of isomorphism classes of vector bundles by

$$\text{Vect}_n^F(X) = \frac{\{p : E \rightarrow B \mid E \text{ is a vector bundle of } B \text{ over } F\}}{\sim}$$

where $E \sim E'$ if E and E' are isomorphic bundles.

1.2 The Cocycle Conditions

Given two charts (U_α, ϕ_α) and (U_β, ϕ_β) of a vector bundle,

$$\phi_\beta \circ \phi_\alpha^{-1} : (U_\alpha \cap U_\beta) \times F^k \rightarrow (U_\alpha \cap U_\beta) \times F^k$$

is a well defined function. In particular, by fixing a point in $U_\alpha \cap U_\beta$, we obtain a linear map.

Definition 1.2.1 (Transition Functions)

Let $p : E \rightarrow B$ be an F -vector bundle of rank k . Let (U_α, ϕ_α) and (U_β, ϕ_β) be local trivialization. Define $g_{\alpha, \beta} : U_\alpha \cap U_\beta \rightarrow \text{GL}(n, F)$ by

$$x \mapsto \phi_\beta \circ \phi_\alpha^{-1}(x, -) : F^k \rightarrow F^k$$

In other words, $g_{\alpha, \beta}$ is such that

$$\phi_\beta \circ \phi_\alpha^{-1}(x, v) = (x, g_{\alpha, \beta}(x)v)$$

For $x \in U_\alpha \cap U_\beta$ and $v \in F^k$.

Proposition 1.2.2

Let $p : E \rightarrow B$ be a K -vector bundle of rank k . The transition functions of the vector bundle satisfies the following.

- Cocycle condition: $g_{\gamma,\alpha}(x) \circ g_{\beta,\gamma}(x) \circ g_{\alpha,\beta}(x) = \text{id}$ on $U_\alpha \cap U_\beta \cap U_\gamma$ for all $x \in U_\alpha \cap U_\beta \cap U_\gamma$.
- $g_{\alpha\alpha}(x) = I_r$ for all $x \in U_\alpha$.

Proof

We have

$$\phi_\alpha \circ \phi_\gamma^{-1} \circ \phi_\gamma \circ \phi_\beta^{-1} \circ \phi_\beta \circ \phi_\alpha^{-1} = \text{id}$$

Hence

$$(x, (g_{\gamma,\alpha}(x) \circ g_{\beta,\gamma}(x) \circ g_{\alpha,\beta}(x))(v)) = (x, v)$$

for all $x \in U_\alpha \cap U_\beta \cap U_\gamma$ and all $v \in F^k$. Hence the cocycle condition is achieved.

Clearly $g_{\alpha,\alpha}(x, -)$ is the identity by definition. ■

The cocycle conditions completely determine a vector bundle.

Definition 1.2.3 (Associated Vector Bundle)

Let F be a field. Let B be a space. Let $\{U_i \mid i \in I\}$ be an open cover of B . Let $g_{i,j} : U_i \cap U_j \rightarrow \text{GL}(k, F)$ be a continuous functions for $i, j \in I$ such that $g_{k,i}(x) \circ g_{j,k}(x) \circ g_{i,j}(x) = \text{id}$ for all $x \in U_i \cap U_j \cap U_k$ and $g_{i,i}(x) = \text{id}$ for all $x \in U_i$. Define the associated vector bundle to be the space

$$E_g = \frac{\coprod_{i \in I} U_i \times F^k}{\sim}$$

where $(i, b, v) \sim (j, b, g_{i,j}(b)(v))$ if $b \in U_i \cap U_j$, together with the induced projection map $\text{proj} : E_g \rightarrow B$.

Lemma 1.2.4

Let F be a field. Let B be a space. Let $\{U_i \mid i \in I\}$ be an open cover of B . Let $g_{i,j} : U_i \cap U_j \rightarrow \text{GL}(k, F)$ be a continuous functions for $i, j \in I$. Then $p : E_g \rightarrow B$ is a vector bundle of rank k .

Proof

Firstly note that the projection map $E_g \rightarrow B$ is well defined by the universal property of quotient. We then need to prove that the local triviality condition is satisfied. Clearly the composition $U_i \times F^k \hookrightarrow \coprod_{i \in I} U_i \times F^k \rightarrow E_g$ composed with the canonical projection map commutes. It is a homeomorphism following the universal property of quotients. ■

Lemma 1.2.5

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Let $g = \{g_{i,j} \mid i, j \in I\}$ be the transition functions. Then the transition functions induce an isomorphism

$$E_g \cong E$$

of vector bundles.

1.3 Operations on Vector Bundles

Definition 1.3.1 (Whitney Sum) Let $p_1 : E_1 \rightarrow B$ and $p_2 : E_2 \rightarrow B$ be two vector bundles. Define the direct sum of the vector bundles to be

$$E_1 \oplus E_2 = \{(v_1, v_2) \in E_1 \times E_2 \mid p_1(v_1) = p_2(v_2)\}$$

together with the projection $p : E_1 \oplus E_2 \rightarrow B$ defined by $(v_1, v_2) \mapsto p_1(v) = p_2(v)$.

Example 1.3.2

Let $p : M \rightarrow S^1$ be the Mobius vector bundle. Then $M \oplus M \cong e^2$.

Proof

Define a morphism $M \oplus M \rightarrow e^2$ by $([(t, x)], [(t, y)]) \mapsto (e^{2\pi i t}, x, y)$. Notice that that restricted to the open cover $U_1 = \{e^{2\pi i t} \mid 0 \leq t \leq 1/2\}$ and $U_2 = \{e^{2\pi i t} \mid 1/2 \leq t \leq 1\}$ of S^1 , we see that the local trivialization charts of $M \oplus M$ coincides with that of our morphism. Hence the morphism is a well defined morphism of vector bundles. Moreover, it is an isomorphism of vector spaces on each fiber. ■

Definition 1.3.3 (Pullback Bundle)

Let F be a field. Let X, Y be spaces. Let $p : E \rightarrow Y$ be a vector bundle over F . Let $f : X \rightarrow Y$ be a map. Define the pullback bundle of f to be the space

$$f^*(E) = \{(x, e) \in X \times E \mid f(x) = p(e)\}$$

together with the projection map $p^* : f^*(E) \rightarrow X$ given by $p^*(x, e) = x$.

Definition 1.3.4 (The Hom Bundle)

Let F be a field. Let B be a space. Let $p_1 : E_1 \rightarrow B$ and $p_2 : E_2 \rightarrow B$ be vector bundles over F . Define the Hom bundle of E_1 and E_2 to be the space

$$\text{Hom}(E_1, E_2) = \coprod_{b \in B} \text{Hom}_F(p_1^{-1}(b), p_2^{-1}(b))$$

together with the coproduct of the maps $\text{Hom}_F(p_1^{-1}(b), p_2^{-1}(b)) \rightarrow \{b\}$.

Definition 1.3.5 (The Dual Bundle)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Define the dual bundle of p to be the vector bundle

$$E^* = \text{Hom}(E, e^1)$$

Definition 1.3.6 (The Tensor Product Bundle)

Let F be a field. Let B be a space. Let $p_1 : E_1 \rightarrow B$ and $p_2 : E_2 \rightarrow B$ be vector bundles over F . Define the tensor product bundle to be the space

$$E_1 \otimes E_2 = \coprod_{b \in B} p_1^{-1}(b) \otimes_F p_2^{-1}(b)$$

together with the coproduct of the maps $p_1^{-1}(b) \otimes p_2^{-1}(b) \rightarrow \{b\}$.

Definition 1.3.7 (The Exterior Product Bundle)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle. Define the n th exterior product bundle of E to be

$$\Lambda^n(E) = \coprod_{b \in B} \Lambda^n(p^{-1}(b))$$

together with the coproduct of the maps $\Lambda^n(p^{-1}(b)) \rightarrow \{b\}$.

1.4 Homotopy Theory of Vector Bundles

Theorem 1.4.1 (The Homotopy Theorem)

Let X be a paracompact space. Let $p : E \rightarrow X \times I$ be a vector bundle. Then there is a bundle isomorphism

$$p^{-1}(X \times \{0\}) \cong p^{-1}(X \times \{1\})$$

Proposition 1.4.2

Let X, Y be paracompact spaces. Let $p : E \rightarrow Y$ be a vector bundle. Let $f, g : X \rightarrow Y$ be homotopic maps. Then there is an isomorphism

$$f^*(E) \cong g^*(E)$$

Corollary 1.4.3

Let X, Y be paracompact spaces. Let $f : X \rightarrow Y$ be a map. If f is a homotopy equivalence, then the pullback map induces natural bijection

$$f^* : \text{Vect}_n^F(Y) \xrightarrow{1:1} \text{Vect}_n^F(X)$$

Corollary 1.4.4

Let B be a paracompact space. Let $p : E \rightarrow B$ be a vector bundle. If B is contractible, then E is trivial.

1.5 Almost Complex Structures

Definition 1.5.1 (Linear Complex Structure)

Let V be a vector space over $\mathbb{R}$. A linear complex structure on V is a linear map $J : V \rightarrow V$ such that $J^2 = -\text{id}_V$.

Proposition 1.5.2

Let V be a vector space over $\mathbb{R}$. Let J be a linear complex structure on V . Then V is a $\mathbb{C}$ -vector space whose action is given by

$$(x + iy) \cdot v = xv + yJ(v)$$

for $v \in V$ and $x + iy \in \mathbb{C}$.

Proposition 1.5.3

Let V be a finite dimensional vector space over $\mathbb{R}$. Then the following are true.

- V admits a linear complex structure if and only if $\dim_{\mathbb{R}}(V)$ is even.
- In this case, we have $\dim_{\mathbb{C}}(V) = 2 \dim_{\mathbb{R}}(V)$.

Example 1.5.4

Let $J : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ be the map defined in block matrix form

$$J = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}$$

Then J defines a linear complex structure on $\mathbb{R}^{2n}$. In particular, if $e_1, \dots, e_{2n}$ are the standard basis vectors for $\mathbb{R}^{2n}$, then $\frac{1}{2}(e_1 - ie_{2n+1}), \dots, \frac{1}{2}(e_n - ie_{2n})$ is a basis for V as a $\mathbb{C}$ -vector space.

Proposition 1.5.5

Let V be a vector space over $\mathbb{R}$. Let J be a linear complex structure on V . Let $T : V \rightarrow V$ be an $\mathbb{R}$ -linear transformation. Then T is a $\mathbb{C}$ -linear transformation if and only if $J \circ T = T \circ J$.

The linear complex structure on an $\mathbb{R}$ -vector space gives an intrinsic way of giving the $\mathbb{R}$ -vector space the structure of a $\mathbb{C}$ -vector space. However, we already know an external way of giving $\mathbb{R}$ -vector spaces $\mathbb{C}$ -multiplication: namely by extension of scalars.

Proposition 1.5.6

Let V be a vector space over $\mathbb{R}$. Let J be a linear complex structure on V . Then the following are true.

- J extends to a linear complex structure on $V \otimes_{\mathbb{R}} \mathbb{C}$ by $J(v \otimes z) = J(v) \otimes z$.

- The eigenvalues of $J : V \otimes_{\mathbb{R}} \mathbb{C} \rightarrow V \otimes_{\mathbb{R}} \mathbb{C}$ is given by $\pm i$ and the eigenspaces of J are given by

$$E_i = \{v \otimes 1 - J(v) \otimes i \mid v \in V\} \quad \text{and} \quad E_{-i} = \{v \otimes 1 + J(v) \otimes i \mid v \in V\}$$

- There is a natural isomorphism of $\mathbb{C}$ -vector space $V \cong E_i$ given by $v \mapsto v \otimes 1 - J(v) \otimes i$.
- The conjugation map $\overline{(-)} : V \otimes_{\mathbb{R}} \mathbb{C} \rightarrow V \otimes_{\mathbb{R}} \mathbb{C}$ given by $\overline{v \otimes z} = v \otimes \bar{z}$ restricts to an isomorphism

$$\overline{(-)}|_{E_i} : E_i \xrightarrow{\cong} E_{-i}$$

whose inverse is given by $\overline{(-)}|_{E_{-i}}$.

Proposition 1.5.7

Let V be a vector space over $\mathbb{R}$. Let J be a linear complex structure on V . Then J^T defines a linear complex structure on V^* .

Definition 1.5.8 (Almost Complex Structure)

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over $\mathbb{R}$. An almost complex structure on p is a vector bundle isomorphism $j : E \rightarrow E$ such that $j \circ j = \text{id}$.

Proposition 1.5.9

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over $\mathbb{R}$. Let J be an almost complex structure on p . Then the following are true.

- The rank of E over $\mathbb{R}$ is even.
- E is a complex vector bundle whose action is given by

$$(x + iy) \cdot v = xv + yJ_b(v)$$

for $v \in E_b$ and $x + iy \in \mathbb{C}$ for each $b \in B$.

Proposition 1.5.10

Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over $\mathbb{C}$ of rank n . Then E is a real vector bundle of rank $2n$.

2 Sequences of Vector Bundles

2.1 Subbundles and Quotient Bundles

Definition 2.1.1 (Vector Subbundles)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . A subbundle of E is a subset $E' \subseteq E$ such that $p|_{E'} : E' \rightarrow B$ is a vector bundle over F .

Lemma 2.1.2

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Let E' be a subbundle of E . Let $\phi_i : p^{-1}(U_i) \rightarrow U_i \times F^k$ be a local chart of E' . Then the following are true.

- There exists a map $\psi_i : p^{-1}(U_i) \rightarrow U_i \times F^{n-k}$ such that the map $p^{-1}(U_i) \rightarrow U_i \times F^n$ given by $x \mapsto (x, \text{proj}_{F^k}(\phi_i(x)), \text{proj}_{F^{n-k}}(\psi_i(x)))$ is a local chart of E .
- The transition functions of E are given in block matrix

$$g_{ij} = \begin{pmatrix} g'_{ij} & l_{ij} \\ 0 & k_{ij} \end{pmatrix}$$

for some maps $l_{ij} : U_i \cap U_j \rightarrow M_{k,n-k}(F)$ and $k_{ij} : U_i \cap U_j \rightarrow \text{GL}(n-k, F)$.

- The maps k_{ij} satisfy the cocycle condition and $k_{ii}(x) = I_{n-k}$ for all $x \in U_i$.

Definition 2.1.3 (The Quotient Bundle)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Let E' be a subbundle of E . Define the quotient bundle E/E' to be the associated vector bundle to the collection of functions k_{ij} given by the above.

Lemma 2.1.4

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Let E' be a subbundle of E . There are linear isomorphisms

$$(E/E')_b \cong E_b/E'_b$$

for all $b \in B$.

Proposition 2.1.5 (Universal Property of Quotient Bundles)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle over F . Let E' be a subbundle of E . Then the quotient bundle and the quotient map $q : E \rightarrow E/E'$ satisfies the following universal property. For any vector bundle G over F and any morphism $f : E \rightarrow G$ such that $f(E')$ is the trivial bundle, there exists a unique morphism $\tilde{f} : E/E' \rightarrow G$ such that the following diagram commutes:

$$\begin{array}{ccc} E & \xrightarrow{q} & E/E' \\ & \searrow f & \downarrow \tilde{f} \\ & & G \end{array}$$

Moreover, E/E' is the unique (up to unique isomorphism) vector bundle that satisfies the above.

2.2 The Kernel of a Morphism

2.3 Short Exact Sequences of Vector Bundles

Definition 2.3.1 (Short Exact Sequences of Vector Bundles)

Let F be a field. Let B be a space. Let E, G, H be vector bundles over B . Let $f : E \rightarrow G$ and $g : G \rightarrow H$ be morphisms. We say that the sequence

$$0 \longrightarrow E \xrightarrow{f} G \xrightarrow{g} H \longrightarrow 0$$

is a short exact sequence of vector bundles if the induced linear maps

$$0 \longrightarrow E_b \xrightarrow{f|_{E_b}} G_b \xrightarrow{g|_{E_b}} H_b \longrightarrow 0$$

are short exact sequences of vector spaces for all $b \in B$.

Lemma 2.3.2

Let F be a field. Let B be a space. Suppose that the following is a short exact sequence of vector bundles over B :

$$0 \longrightarrow E \xrightarrow{f} G \xrightarrow{g} H \longrightarrow 0$$

Then the following are true.

- g induces an isomorphism $G/E \cong H$.
- f induces an isomorphism $\ker(G \rightarrow H) \cong E$.

TBA: Exactness of taking sections TBA: Exactness of tensor products

Note: category of vector bundles is not an abelian category.

3 Reduction of Structure Group

3.1 Structure Groups

3.2 Metrics and Reduction to Orthogonal Groups

Definition 3.2.1 (Metrics)

Let F be a field. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle.

- Suppose that $F = \mathbb{R}$. A metric on E is a smooth section of $s \in (E \otimes E)^* \cong E^* \otimes E^*$ such that $s_p : E_b \times E_b \rightarrow \mathbb{R}$ is an inner product for all $b \in B$.
- Suppose that $F = \mathbb{C}$. A metric on E is a smooth section of $s \in (E \otimes E)^* \cong E^* \otimes E^*$ such that $s_p : E_b \times E_b \rightarrow \mathbb{R}$ is a Hermitian product for all $b \in B$.

Lemma 3.2.2

Let $F = \mathbb{R}$ or $\mathbb{C}$. Let B be a space. Let $p : E \rightarrow B$ be a vector bundle. If B is paracompact, then E admits a metric.

Note: ses of vector bundles of paracompact spaces are split exact. Note: vector bundles have reduction of structure group to $O(n)$ and $U(n)$ if and only if they admit a metric.

3.3 Orientability and Reduction to Special Linear Groups

vector bundle admits orientation if and only they have reduction of structure group to $SL(n, \mathbb{R})$ and $SL(n, \mathbb{C})$.

3.4 Subbundles and Reduction to Product Orthogonal Subgroups

$O(n)$ -vector bundle admits k -dimensional subbundle if and only if it has reduction of structure group to $O(k) \times O(n - k)$.

4 The Topology of Fiber Bundles

4.1 Fiber Bundles

Fiber bundles serve as somewhat of a generalization of both vector bundles and covering spaces, while being a special case of a fibration. It therefore has the properties of a fibration.

Definition 4.1.1 (Fiber Bundles)

Let E, B, F be spaces with B connected, and $p : E \rightarrow B$ a continuous map. We say that p is a fiber bundle over F if the following are true.

- $p : E \rightarrow B$ is surjective
- Local Triviality: There exists a collection

$$\{(U_k \subseteq B, \phi_k : p^{-1}(U_k) \xrightarrow{\cong} U_k \times F) \mid k \in I\}$$

called local trivialization charts such that the following diagram commutes:

$$\begin{array}{ccc} p^{-1}(U_k) & \xrightarrow{\phi_k} & U_k \times F \\ & \searrow p & \swarrow \pi \\ & U_k & \end{array}$$

where π is the projection by forgetting the second variable.

We say that B is the base space, E the total space. It is denoted as (F, E, B) or $F \hookrightarrow E \rightarrow B$.

Intuitively, we would like a fiber bundle to locally look like the product $B \times F$.

Lemma 4.1.2

Let K be a field. Let $p : E \rightarrow B$ be a fiber bundle with fiber F . Then the local trivialization charts induces a homeomorphism

$$p^{-1}(b) \cong F$$

for all $b \in B$.

Fiber bundles generalize vector bundles in the sense that the fibers are no longer vector spaces but instead arbitrary spaces.

Lemma 4.1.3

The following are true regarding fiber bundles.

- Every vector bundle is a fiber bundle.
- Every covering space is a fiber bundle.

Proof Indeed if $p : E \rightarrow B$ is a vector bundle, then each fiber $p^{-1}(b)$ is an n -dimensional vector spaces over a field F . Moreover, by definition the local triviality condition is also satisfied.

Finally, for any $U \subseteq X$, recall that

$$p^{-1}(U) = \coprod_{i \in I} V_i$$

where each $V_i \cong U$. It is clear by definition that $|p^{-1}(x)| = |I|$ for any $x \in X$. By giving I the discrete topology, we obtain a homeomorphism $p^{-1}(x) \cong I$. The homeomorphism $p^{-1}(U) = \coprod_{i \in I} V_i$ translates to

$$p^{-1}(U) = \coprod_{i \in I} V_i \cong \coprod_{i \in I} U \cong U \times I$$

defined by $\tilde{x} \in V_i \mapsto (p(\tilde{x}) = x, i)$. It is thus clear that the local triviality condition is satisfied. ■

Example 4.1.4

The following maps are fiber bundles.

- Let $f_n : S^1 \rightarrow S^1$ be the map defined by $f_n(z) = z^n$. Then f_n is a fiber bundle with fiber the

n th roots of unity.

- Let $f : \mathbb{R} \rightarrow S^1$ be the map defined by $f(t) = e^{2\pi it}$. Then f is a fiber bundle with fiber $\mathbb{Z}$.
- Let $p : S^n \rightarrow \mathbb{RP}^n$ be the projection map. Then p is a fiber bundle with fiber as a two point set with the discrete topology.

Proof

Indeed, all of these are covering spaces and hence fiber bundles by the above lemma. ■

Example 4.1.5

Let $M = \frac{[-1,1] \times [-1,1]}{(t,-1) \sim (1-t,1)}$ be the Mobius strip. Then the projection map $p : M \rightarrow C$ defined by $p(t, s) = s$ is a fiber bundle with fiber $[-1, 1]$.

Proposition 4.1.6 Every fiber bundle is a Serre fibration.

We can provide a partial converse for the fact that every fiber bundle is a Serre fibration.

Proposition 4.1.7 Let $p : E \rightarrow B$ be a fiber bundle. If B is paracompact, then p is a (Hurewicz) fibration.

We there fore have inclusions

$$\text{Fiber Bundles} \subset \text{Serre Fibrations} \subset \text{(Hurewicz) Fibrations}$$

Definition 4.1.8 (Map of Fiber Bundles)

Let (F_1, E_1, B_1) and (F_2, E_2, B_2) be fiber bundles. A map of fiber bundles is a pair of basepoint preserving continuous maps $(\tilde{f} : E_1 \rightarrow E_2, f : B_1 \rightarrow B_2)$ such that the following diagram commutes:

$$\begin{array}{ccc} E_1 & \xrightarrow{\tilde{f}} & E_2 \\ p_1 \downarrow & & \downarrow p_2 \\ B_1 & \xrightarrow{f} & B_2 \end{array}$$

A map of fiber bundles $(\tilde{f}, f)$ is said to be an isomorphism if there is a map $(\tilde{g} : E_2 \rightarrow E_1, g : B_2 \rightarrow B_1)$ such that $\tilde{g}$ is the inverse of $\tilde{f}$ and g is the inverse of f .

Such a map of fiber bundles determine a continuous of the fibers $F_1 \cong p_1^{-1}(b_1) \rightarrow p_2^{-1}(b_2) \cong F_2$. Moreover, such a morphism preserves fibers. Indeed, If $p_1^{-1}(b)$ is a fiber of B , then using the commutativity of the diagram we have that

$$p_2(\tilde{f}(p_1^{-1}(b))) = f(p_1(p_1^{-1}(b))) = f(b)$$

which implies that

$$p_2^{-1}(f(b)) = \tilde{f}(p_1^{-1}(b))$$

or in other words, the fiber at $f(b)$ is the same as the fiber at b applied with $\tilde{f}$.

Definition 4.1.9 (Isomorphic Fiber Bundles)

Let $p_1 : E_1 \rightarrow B_1$ and $p_2 : E_2 \rightarrow B_2$ be two fiber bundles. Let $(\tilde{f}, f) : p_1 \rightarrow p_2$ be a map of fiber bundles. We say that $(\tilde{f}, f)$ is an isomorphism if there exists a map $(\tilde{g}, g) : p_2 \rightarrow p_1$ of fiber bundles such that $\tilde{f}$ is the inverse of $\tilde{g}$ and f is the inverse of g . In this case, we say that p_1 and p_2 are isomorphic.

There are two important special cases of fiber bundles that will appear time and time again.

Definition 4.1.10 (Trivial Bundles)

We say that a fiber bundle (F, E, B) is trivial if (F, E, B) is isomorphic to the trivial fibration $B \times F \rightarrow B$.

Definition 4.1.11 (The Pullback Bundle) Let $p : E \rightarrow B$ be a fiber bundle with fiber F . Let $f : B' \rightarrow B$ be a continuous function. Define the pullback of p by f to be the space

$$f^*(E) = \{(b', e) \in B' \times E \mid p(e) = f(b')\}$$

Theorem 4.1.12 Let $p : E \rightarrow B$ be a fiber bundle. Suppose that $f, g : X \rightarrow B$ are homotopic maps. Then the pull back bundles

$$f^*(E) \cong g^*(E)$$

are equivalent.

4.2 Sections of a Bundle

Definition 4.2.1 (Sections) Let (F, E, B) be a fiber bundle. A section on the fiber bundle is a map $s : B \rightarrow E$ such that

$$p \circ s = \text{id}_B$$

Definition 4.2.2 (Local Sections) Let (F, E, B) be a fiber bundle. Let $U \subset B$ be an open set. A local section of the fiber bundle on U is a map $s : U \rightarrow E$ such that

$$p \circ s = \text{id}_U$$

4.3 Sphere Bundles

We now consider a special type of fibrations where the fibers are given by S^1 . We define a class of fiber bundles of the form $S^1 \rightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$. When $n = 1$, this is the Hopf fibration.

Definition 4.3.1 (Sphere Bundles) A sphere bundle is a fiber bundle $p : E \rightarrow B$ for which its fibers are the n -sphere S^n .

Example 4.3.2

Let $n \in \mathbb{N}$. Consider S^{2n+1} lying inside $\mathbb{C}^{n+1}$. The projection map $\mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{CP}^n$ induces a fiber bundle

$$S^1 \hookrightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$$

Proof

Clearly the projection map $p : S^{2n+1} \rightarrow \mathbb{CP}^n$ is surjective. Consider the open cover $\{U_i \mid 0 \leq i \leq n\}$ where $U_i = \{[z_0 : \dots : z_n] \mid z_i \neq 0\}$. Define the local trivialization charts $\phi_i : p^{-1}(U_i) \rightarrow U_i \times S^1$ by $(z_0, \dots, z_n) \rightarrow ([z_0 : \dots : z_n], z_i/|z_i|)$. It is continuous since it is the product of the composition map and the projection map to the sphere S^1 . Its inverse is the continuous map $([z_0, \dots, z_n], w) = w|z_i|z_i^{-1}(z_0, \dots, z_n)$ since it is induced by the universal property of quotients. It is clear that the required diagram commutes, that is $p = \text{proj} \circ \phi_i$ where $\text{proj} : U_i \times S^1 \rightarrow U_i$ is the projection map to the first coordinate. ■

Example 4.3.3

For each $n \in \mathbb{N}$, we have a map of fiber bundles:

$$\begin{array}{ccc} S^{2n+1} & \hookrightarrow & S^{2n+3} \\ \downarrow & & \downarrow \\ \mathbb{CP}^n & \hookrightarrow & \mathbb{CP}^{n+1} \end{array}$$

whose horizontal maps are induced by the inclusion $\mathbb{C}^n \hookrightarrow \mathbb{C}^{n+1}$ into the first n coordinates. Taking colimit of the diagram, we have that

$$S^1 \hookrightarrow S^\infty \rightarrow \mathbb{CP}^\infty$$

is a fiber bundle.

Proof

Since the local triviality condition is a local condition, for each $x \in \mathbb{CP}^\infty$, x lives in $\mathbb{CP}^n$ for some $n \in \mathbb{N}$ and since $S^1 \hookrightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$ is a fiber bundle, there exists a chart for which the local triviality condition is satisfied. ■

Example 4.3.4 (Hopf Fibration / Hopf Bundle)

Consider the above example with $n = 1$. Restricting the total space to S^3 , the fiber bundle $p : S^3 \rightarrow S^2$ with fiber S^1 is called the Hopf fibration / Hopf bundle.